AR Lecture 2

Introduction to Virtual Reality

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Definition of Virtual Reality

- Virtual Reality (VR) is a computer-generated **3D simulated environment** that allows users to **experience and interact** with a virtual world in real time using specialized hardware and software.

Key Characteristics of VR:

- Computer-generated environment
- Real-time interaction
- Multi-sensory feedback (visual, audio, haptic)
- User immersion

Core Fundamentals of Virtual Reality

A. Immersion

- **Definition:**
  Immersion refers to the **degree to which a user feels surrounded by the virtual environment**.
- **Factors affecting immersion:**
  - Field of View (FOV)
  - Frame rate
  - Display resolution
  - Tracking accuracy
  - Audio realism

- **Field of View (FOV)**
  Field of View in VR refers to the **extent of the virtual world visible to the user at any given moment**, measured in degrees.

- **Frame Rate**
  Frame rate is the **number of frames displayed per second (FPS)** in a VR system.

- Display Resolution

Display resolution refers to the **number of pixels used to display the VR image**, usually expressed as width × height per eye.

- Tracking Accuracy

Tracking accuracy is the **precision with which a VR system detects and updates the user’s head, hand, or body movements** in real time.

- Audio Realism

Audio realism in VR refers to the **use of spatial and 3D sound to simulate real-world audio perception**.

B. Presence

- **Definition:**
  Presence is the **psychological sensation of “being there”** inside the virtual environment.
- **Types of Presence:**
  – Spatial presence
  – Social presence
  – Self presence

1. Spatial Presence

Spatial presence in VR is the **feeling of physically being inside the virtual environment**, rather than in the real world.

2. Social Presence

Social presence in VR is the **sense of being with other real or virtual entities** within the virtual environment.

3. Self Presence

Self presence in VR is the **degree to which users identify their virtual avatar or body as themselves**.

C. Interaction (10 minutes)

Definition:

Interaction is the ability of the user to **manipulate and respond to virtual objects and environments**.

- **Interaction Methods:**
  - Hand controllers
  - Gesture tracking
  - Eye tracking
  - Voice commands

Types of Virtual Reality

- **Non-Immersive VR**
  - **Description:**
  - Desktop-based VR
  - Interaction via keyboard, mouse, joystick
- **Characteristics:**
  - Low immersion
  - Low cost
  - Limited interaction
- **Examples:**
  - Computer-based simulations
  - 3D games on monitors
- **Applications:**
  - Education
  - Training simulations

B. Semi-Immersive VR

- **Description:**
  - Partial immersion
  - Large screens or projection systems
- **Characteristics:**
  - Moderate immersion
  - Uses tracking systems
- **Examples:**
  - Flight simulators
  - CAVE systems
- **Applications:**
  - Engineering training
  - Research labs

C. Fully Immersive VR

- **Description:**
  - Complete immersion
  - Uses VR headsets and sensors
- **Characteristics:**
  - High immersion and presence
  - Real-time interaction
- **Examples:**
  - Meta Quest
  - HTC Vive
- **Applications:**
  - Gaming
  - Medical training
  - Virtual tourism

Comparisons

Feature	Non-Immersive	Semi-Immersive	Fully Immersive
Immersion	Low	Medium	High
Hardware	Monitor	Projection	HMD
Cost	Low	Medium	High
Interaction	Limited	Moderate	Advanced

Recap & Discussion

Key Takeaways:

- Immersion affects realism
- Presence affects psychological experience
- Interaction enables control
- VR systems vary by immersion level